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FILE MEMO

SUBJECT: Length of Serial Division

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Two schemes have been proposed so far:

1. $Q = D - V + 1$
   where Q = no. of digits (bits for binary) in quotient
   D = no. of significant digits in dividend
   V = no. of significant digits in divisor

2. Use the shift designation in the DIVIDE instruction to define the position where the division is to stop.

Rule 1, which was stated in the Harvest Manual, is actually misleading. It purports to fix the quotient length; but the length cannot really be fixed this way because the dividend length may be indeterminate if it resulted from arithmetic operations which permit carry overflow. Moreover, to control the length of the division, the dividend length must be adjusted in advance by a shifting process which is awkward in the B machine.

Rule 2 is the more natural one for the B machine. It does not fix the quotient length. Instead, the shift specifies the rightmost digit of the dividend which enters into the division process. On the last subtraction cycle, the rightmost digit of the divisor lines up with the dividend digit specified by the shift.

Implied in both rules is the ability to carry out division with any non-zero divisor. Having fixed the right end of the quotient by Rule 2, the left end depends on the size of the numbers. The maximum quotient length is the length of the dividend remaining after the shift. The length of the remainder is the length of the divisor plus the number of positions to the right of the shift position in the dividend.

Examples using Rule 2

Note:
1. Examples are decimal; binary division is analogous.
2. A decimal point is shown in the dividend immediately to the right of the position specified by the shift. The shift is shown as number of decimal digits.
3. Leading (non-significant) zeros are not shown because they would not alter the division process.

Dividend	Shift	Divisor	Quotient	Remainder	Comment
80000.	0	6	13333	2	
800.00	2	6	133	200	
800.00	2	60	13	2000	
200.00	2	60	3	2000	
80000	Any	0	0	80000	Divide Overflow

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